

CIV102 Project Team 309 Design Calculations

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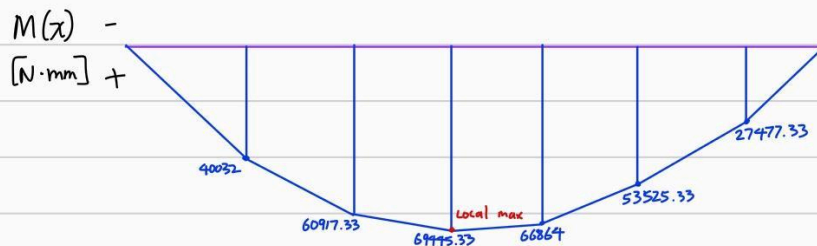
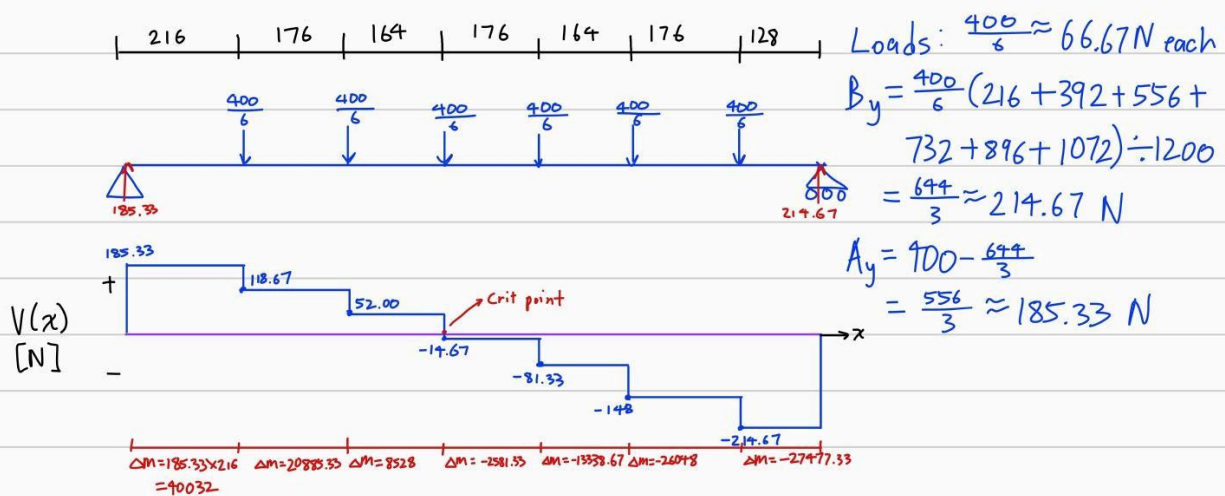
1. Design 0, Load Case 1 - Hand Calculations

Through the MATLAB code that generated the SFE and BME, it was determined that:

$$M_{\max} = 69445.33 \text{ N}\cdot\text{mm}$$

→ Location (From left of bridge): 556 mm

→ Train (from left of bridge to leftmost point load): 216 mm



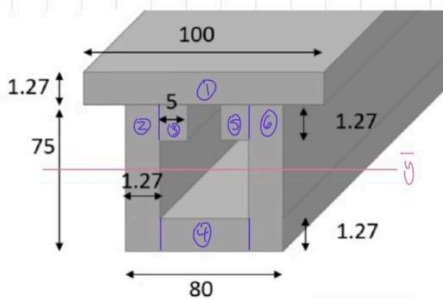
$$\therefore M_{\max} = 69445.33 \text{ N}\cdot\text{mm}$$

$$\therefore x = 216 + 176 + 164 = 556 \text{ mm}$$

Top of beam in compression

Bottom in tension

From Deliverable 1 Calculations:



$$\bar{y} = \frac{\sum (b_i \cdot h_i) \cdot y_i}{\sum (b_i \cdot h_i)} = \frac{(100 \times 1.27)(75 + \frac{1.27}{2}) + 2 \times (75 \times 1.27)(\frac{75}{2}) + 2(5 \times 1.27)(75 - \frac{1.27}{2}) + ((80 - 2 \times 1.27) \times 1.27)(\frac{1.27}{2})}{(100 \times 1.27) + 2(75 \times 1.27) + 2(5 \times 1.27) + ((80 - 2 \times 1.27) \times 1.27)}$$

$$= 41.431 \text{ mm}$$

$$I_x = \frac{bh^3}{12} + A_i d_i^2 \quad \text{distance from } \bar{y}$$

$$I_1 = \frac{100 \times 1.27^3}{12} + (100 \times 1.27)(75 + \frac{1.27}{2} - \bar{y})^2$$

$$= 148546.1$$

$$I_2 = I_3 = \frac{1.27 \times 75^3}{12} + (75 \times 1.27)(\frac{75}{2} - \bar{y})^2$$

$$= 46120.313$$

$$I = \sum_{i=1}^4 I_i = 418352.209 \text{ mm}^4$$

$$I_3 = I_5 = \frac{5 \times 1.27^3}{12} + (5 \times 1.27)(75 - \frac{1.27}{2} - \bar{y})^2$$

$$= 6888.37$$

$$I_4 = \frac{(80 - 2 \times 1.27) \times 1.27^3}{12} + 1.27(80 - 2 \times 1.27)(\frac{1.27}{2} - \bar{y})^2$$

$$= 163738.743$$

$$\bar{y} = 41.431 \text{ mm}$$

$$I = 418352.209 \text{ mm}^4$$

Yielding Failure:

$$\sigma_{\text{applied, comp}} = \frac{M(y_{\text{top}} - \bar{y})}{I} = \frac{69445.33(76.27 - 41.431)}{418352.209} = 5.78 \text{ MPa}$$

$$FOS_{\text{comp}} = \frac{\sigma_c'}{\sigma_{\text{app, c}}} = \frac{6}{5.78} = 1.037$$

$$\sigma_{\text{applied, tens}} = \frac{M \bar{y}}{I} = \frac{69445.33(41.431)}{418352.209} = 6.88 \text{ MPa}$$

$$FOS_{\text{tens}} = \frac{\sigma_t'}{\sigma_{\text{app, t}}} = \frac{30}{6.88} = 4.36$$

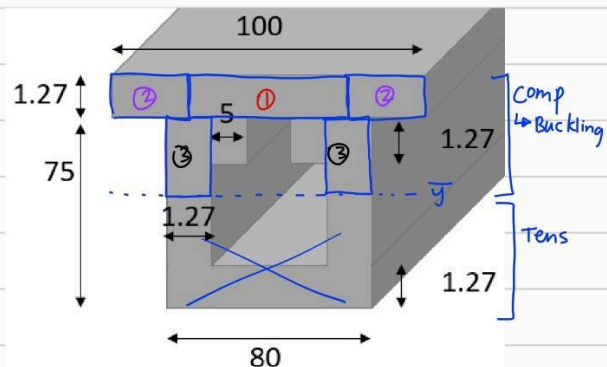
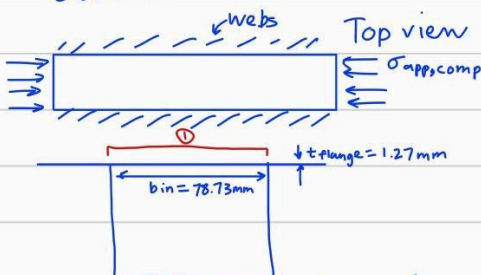
For Matboard:

$$\sigma_c' = 6 \text{ MPa}$$

$$\sigma_t' = 30 \text{ MPa}$$

Compression Buckling Failure:

Case ①:



$$\sigma_{cr,1} = \frac{4\pi^2 E}{12(1 - \nu^2)} \left(\frac{t_{\text{flange}}}{b_{\text{in}}} \right)^2 = \frac{4\pi^2 \cdot 4000}{12(1 - 0.2^2)} \left(\frac{1.27}{78.73} \right)^2 = 3.57 \text{ MPa}$$

$$\sigma_{\text{app, c, top}} = 5.78 \text{ MPa}$$

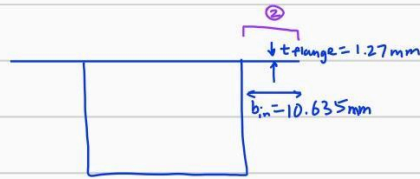
$$FOS_{\text{buckl}} = \frac{3.57}{5.78} = 0.618$$

Matboard:

$$E = 4000 \text{ MPa}$$

$$\nu = 0.2$$

Case ②:

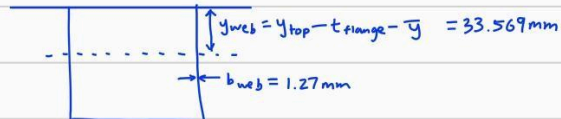
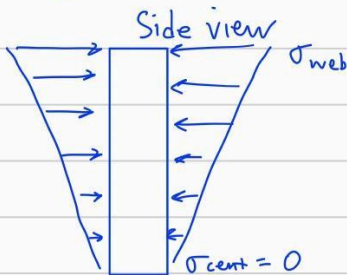


$$\sigma_{cr,2} = \frac{0.4254 \pi^2 E}{12(1-\mu^2)} \left(\frac{t_{flange}}{b_{in}} \right)^2 = \frac{0.4254 \pi^2 \cdot 4000}{12(1-0.2^2)} \left(\frac{1.27}{10.635} \right)^2 = 20.79 \text{ MPa}$$

$$\sigma_{app,c,top} = 5.78 \text{ MPa}$$

$$FOS_{buck2} = \frac{20.79}{5.78} = 3.60$$

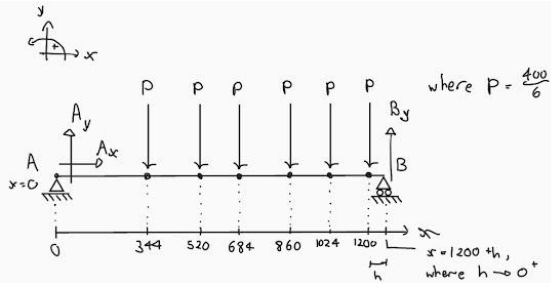
Case ③



$$\sigma_{cr,3} = \frac{6 \pi^2 E}{12(1-\mu^2)} \left(\frac{b_{web}}{y_{web}} \right)^2 = \frac{6 \pi^2 \cdot 4000}{12(1-0.2^2)} \left(\frac{1.27}{33.569} \right)^2 = 29.4 \text{ MPa}$$

$$\sigma_{app,c,web} = \frac{M_{y_{web}}}{I} = \frac{69445.33(33.569)}{418352.209} = 5.57 \text{ MPa}$$

$$FOS_{buck3} = \frac{29.4}{5.57} = 5.28$$



where $P = \frac{400}{6}$

Reaction Forces

$$\sum F_x = 0 \Rightarrow A_x = 0$$

$$\sum M_A = 0 \Rightarrow 0 = B_y(1200 + h)$$

$$-P(344 + 520 + 684 + 860 + 1024 + 1200)$$

$$\Rightarrow B_y = \lim_{h \rightarrow 0} \frac{P(344 + 520 + 684 + 860 + 1024 + 1200)}{1200 + h}$$

$$= 3.86 P, P = \frac{400}{6}$$

$$\Rightarrow B_y = 257.33333 \text{ N}$$

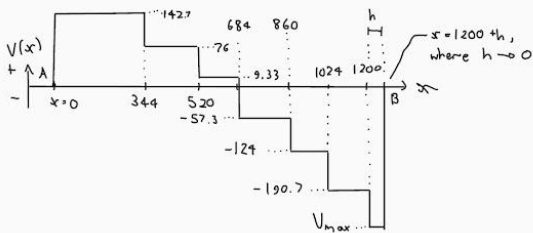
$$\sim 257 \text{ N}$$

$$\sum F_y = 0 \Rightarrow A_y = 6P - B_y, P = \frac{400}{6}, B_y = \frac{3.86(400)}{6}$$

$$\Rightarrow A_y = 142.66667 \text{ N}$$

$$\sim 142.7 \text{ N}$$

Shear Force Diagram

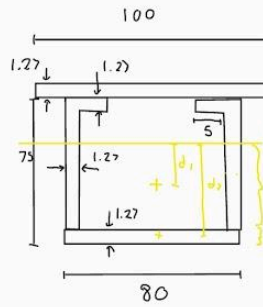


$$\lim_{h \rightarrow 0^+} V_{max} = -B_y = -257.33333 \text{ N}$$

$$\sim -257 \text{ N}$$

Note:

The maximum shear force occurs near the support (point B) when the front wheel is about to reach the support. The shear force decreases when the wheel aligns with the support because the point load and reaction force align. Thus, we let h represent the distance between the front wheel and the support and take the limit of the shear force as h approaches 0 from the positive side.



Shear failure of

walls:

$$\bar{y} = 41.43109435$$

$$A_1 = 2(1.27) \cdot (\bar{y} - 1.27)$$

$$= 102.0097965$$

$$d_1 = (\bar{y} - 1.27)/2$$

$$= 20.09047175$$

$$A_2 = 80 \cdot 1.27$$

$$= 101.6$$

$$d_2 = \bar{y} - (1.27/2)$$

$$= 40.7959435$$

$$Q = A_1 d_1 + A_2 d_2$$

$$\sim 6.20 \text{ mm}^3 \quad \text{mm}^3 = 2.54 \text{ mm}$$

$$\tau = \frac{VQ}{Ib}$$

$$I = 418352.203$$

$$V = 257.3333333$$

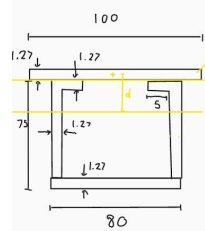
$$\therefore \tau_{web} = 1.499819712 \text{ MPa} \sim 1.500 \text{ MPa}$$

F.O.S. for shear failure in web is

$$\frac{\tau_{max}}{\tau_{web}} = \frac{4 MR}{1.499819712 \text{ MPa}}$$

$$FOS_{web} = 2.666987217 \text{ MPa}$$

$$\approx 2.67 \text{ MPa}$$



Shear failure of

glue:

$$\bar{y} = 41.431$$

$$A = 1.27(100)$$

$$= 127$$

$$d = (75 + \frac{1.27}{2}) - \bar{y}$$

$$= 34.2040565$$

$$Q = A d$$

$$= 127(34.204)$$

$$= 4343.908 \sim 4.34 \times 10^3$$

$$b = 2(5) + 2(1.27) = 12.54$$

$$\tau = \frac{VQ}{Ib}$$

$$I = 418352.203$$

$$V = 257.3333333$$

$$\therefore \tau_{glue} = 0.213077239 \text{ MPa}$$

$$\approx 0.213 \text{ MPa}$$

F.O.S. for shear failure in glue is

$$\frac{\tau_{max}}{\tau_{glue}} = \frac{2 MR}{0.213077239 \text{ MPa}}$$

$$FOS_{glue} = 9.386267657$$

$$\approx 9.39$$

Shear buckling in the web

$$h = 75 - 2(1.27) \quad a = 400$$

$$= 72.46 \text{ mm} \quad t = 1.27$$

$$\mu = 0.2$$

$$E = 4000 \text{ MPa}$$

$$\tau_{buck} = \frac{\pi^2 \cdot E}{12(1-\mu^2)} \left(\left(\frac{t}{h} \right)^2 + \left(\frac{t}{a} \right)^2 \right)$$

$$= \frac{\pi^2 \cdot 4000}{12(1-0.2^2)} \left(\left(\frac{1.27}{72.46} \right)^2 + \left(\frac{1.27}{400} \right)^2 \right)$$

$$= 5.436391401 \approx 5.44 \text{ MPa}$$

$$F.O.S. = \frac{\tau_{max}}{\tau_{web}} = \frac{5.25661985}{1.499819712}$$

$$= 3.507834787$$

$$\approx 3.50$$

2. Design 0, Load Case 1 - Code Output

2.1. Intermediate Calculations & Cross-Section Diagram

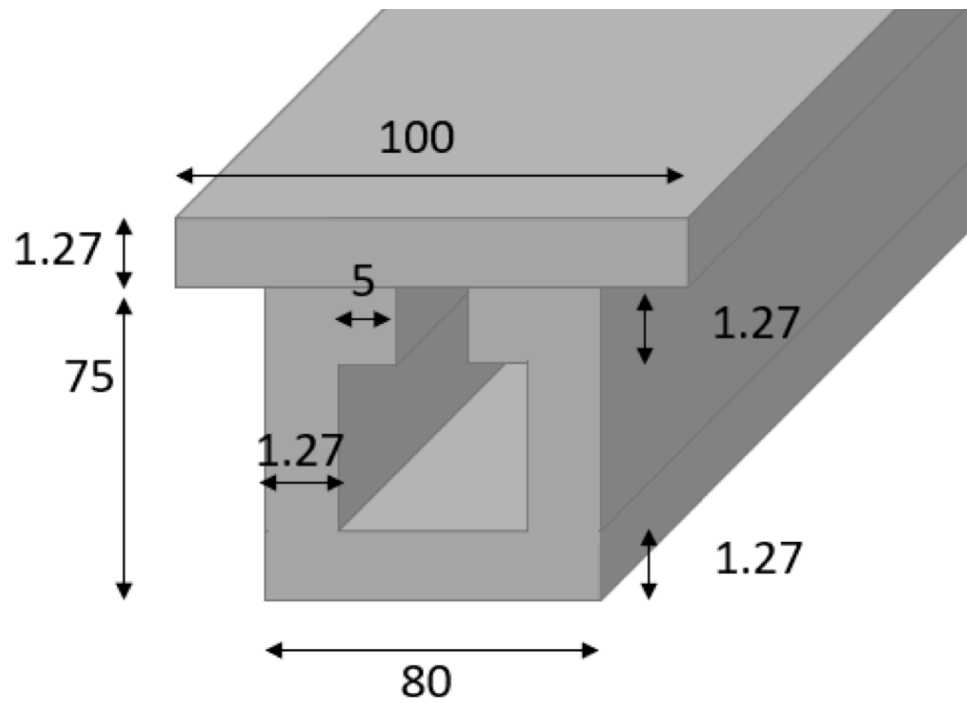


Figure 1: Cross-Section Diagram for Design 0.

Table 1: Intermediate Calculations for Design 0 under Load Case 1.

	Intermediate Calculation	Value
1	Centroidal Axis Location From Bottom (mm)	41.4
2	Second Moment of Area (mm ⁴)	418000
3	Top Bending Stress - Compression (MPa)	5.78
4	Bottom Bending Stress - Tension (MPa)	6.88
5	First Moment of Area at the Centroid (mm ³)	6190
6	Shear Stress at the Centroid (MPa)	1.498
7	First Moment of Area at the Glue Tab (mm ³)	4340
8	Shear Stress at the Glue Tab (MPa)	0.213
9	Shear Buckling Stress (MPa)	5.44
10	Thin Plate Buckling Stress - Top Flange - Middle (MPa)	3.57
11	Thin Plate Buckling Stress - Top Flange - Sides (MPa)	20.8
12	Thin Plate Buckling Stress - Webs (MPa)	29.4

2.2. Shear Force Capacities

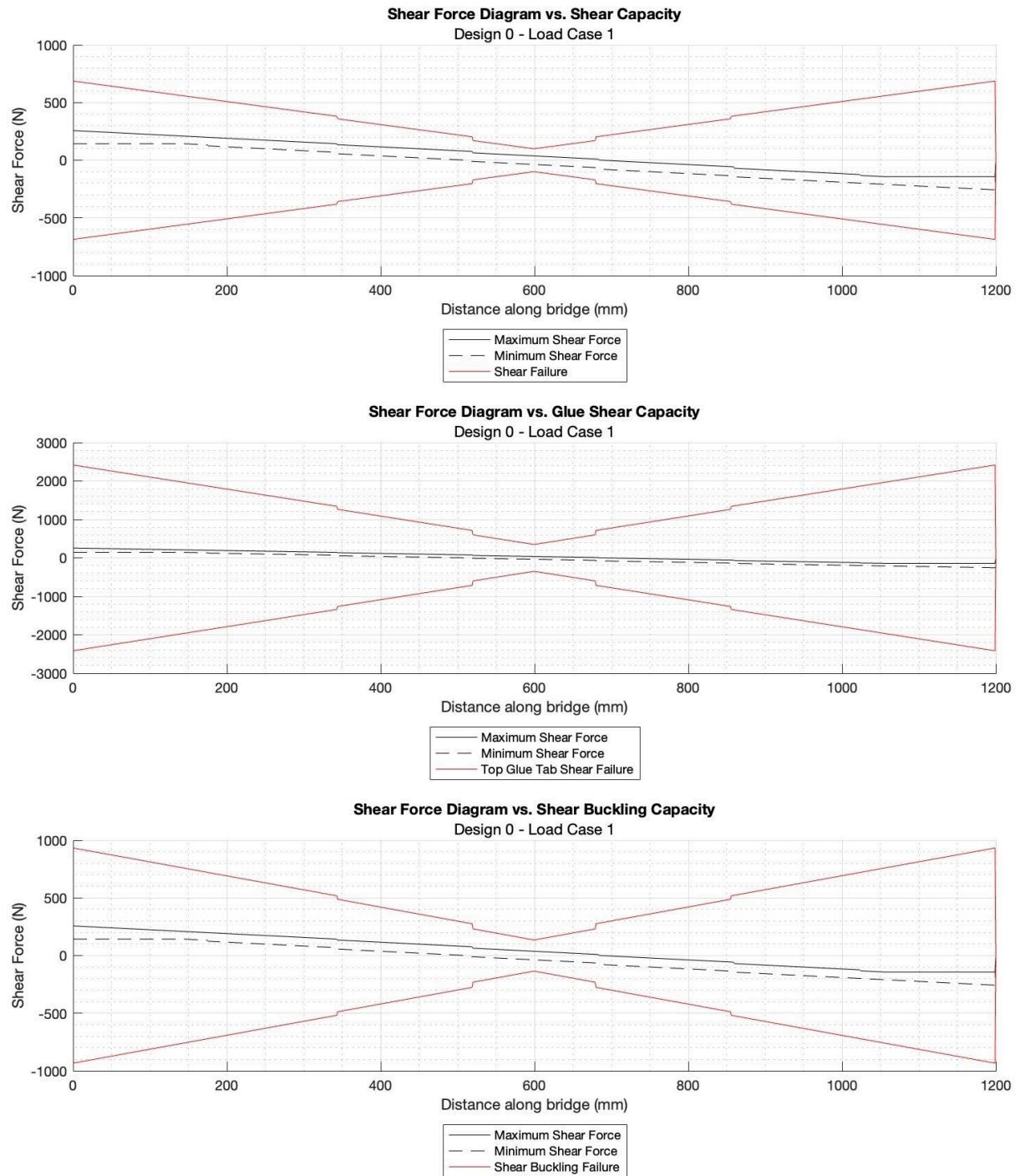


Figure 2: Shear Force Diagram vs. Shear Buckling Capacities for Design 0 under Load Case 1. Closest to shear failure at the centroid. Decently far away from other shear force failure modes.

2.3. Bending Moment Capacities

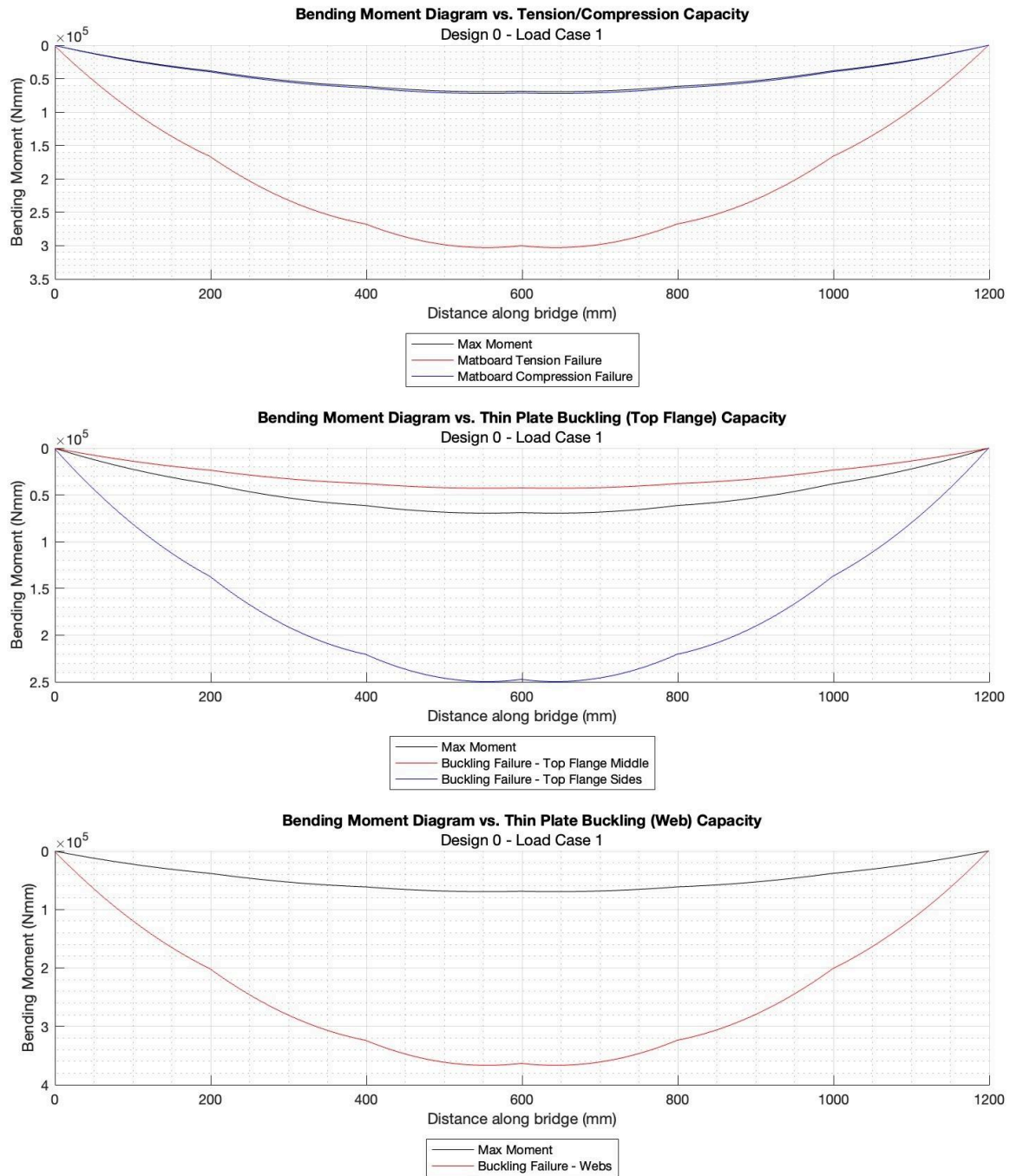


Figure 3: Bending Moment Diagram vs. Bending Moment Capacities for Design 0 under Load Case 1. Fails by buckling failure in the middle of the top flange. Very close to failing by compression. Significantly far from failing by other bending moment failure modes.

2.4. Factors of Safety & Maximum Train Weight

Table 2: Factors of safety against the previous failure modes in Section 2.2. and 2.3. Failure is caused by buckling in the middle of the top flange. This occurs at a maximum train weight of 247 N. The significant figures in the table below are slightly inaccurate because MATLAB does not display trailing zeros. For example, top glue tab shear failure has a FOS of 9.40, which is shown as 9.4, despite requiring three significant figures.

	Failure Mode	FOS
1	Web Shear Failure	2.67
2	Top Glue Tab Shear Failure	9.4
3	Shear Buckling Failure	3.63
4	Tension Failure	4.36
5	Compression Failure	1.037
6	Top Flange Middle Thin Plate Buckling Failure	0.617
7	Top Flange Sides Thin Plate Buckling Failure	3.59
8	Web Thin Plate Buckling Failure	5.28
9		
10	Maximum Train Weight (N)	247

3. Final Design, Load Case 2 - Code Output

3.1. Intermediate Calculations & Cross Section Sketches

3.1.1. Cross Section Case 1: No O-Shape, No Double Bottom

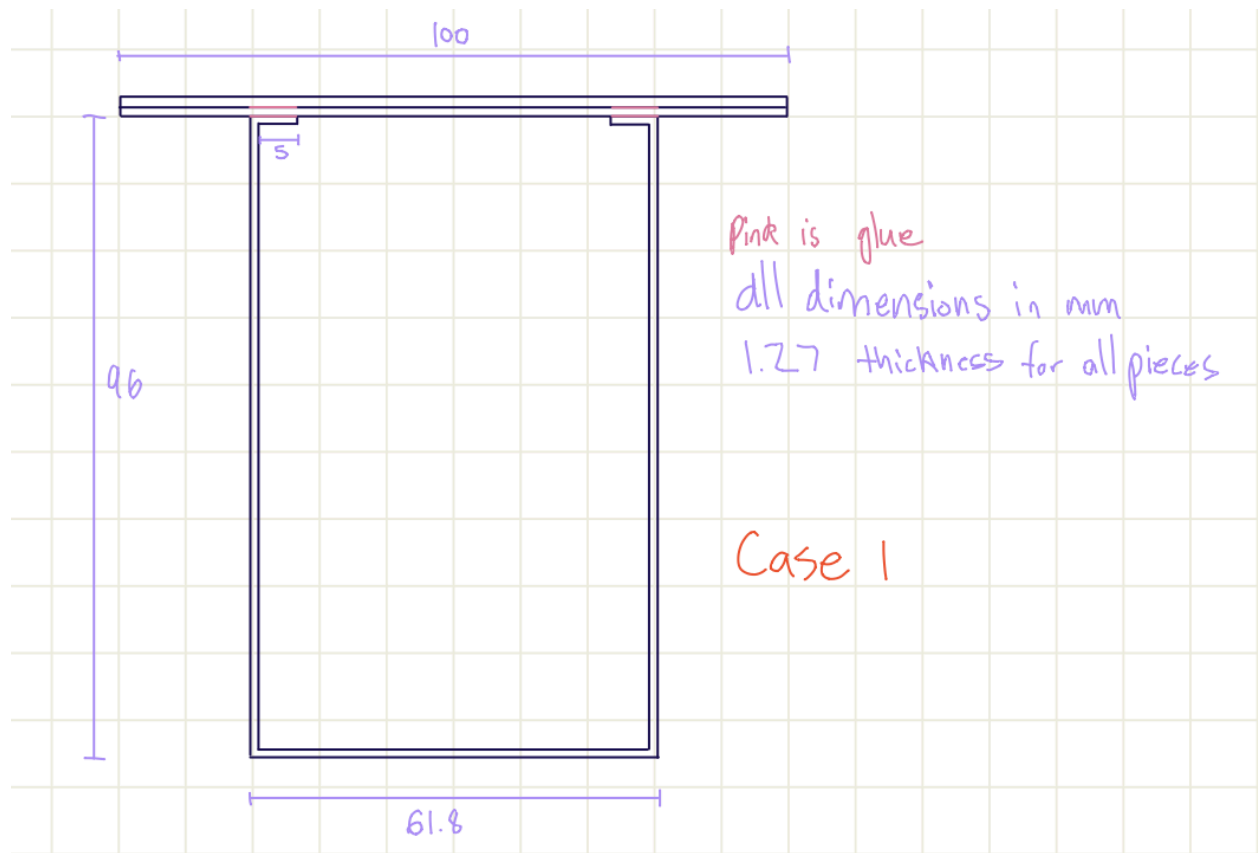


Figure 4: Cross Section Case 1 Diagram. This case is for distances in mm (0, 360) and (900, 1260) along the bridge's span.

3.1.2. Cross Section Case 2: No O-Shape, Yes Double Bottom

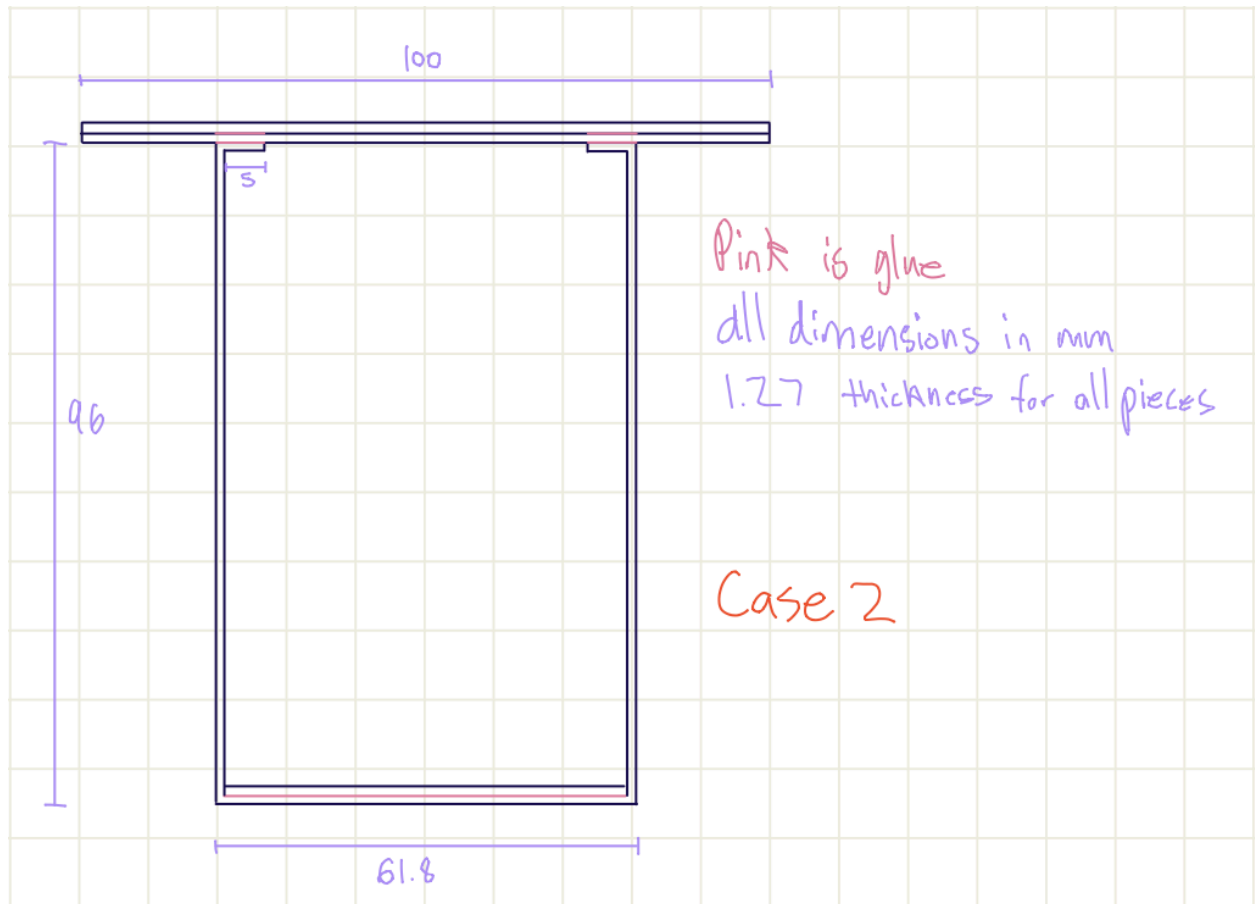


Figure 5: Cross Section Case 2 Diagram. This case is for distances in mm (360, 450) and (810, 900) along the bridge's span.

3.1.3. Cross Section Case 3: Yes O-Shape, Yes Double Bottom

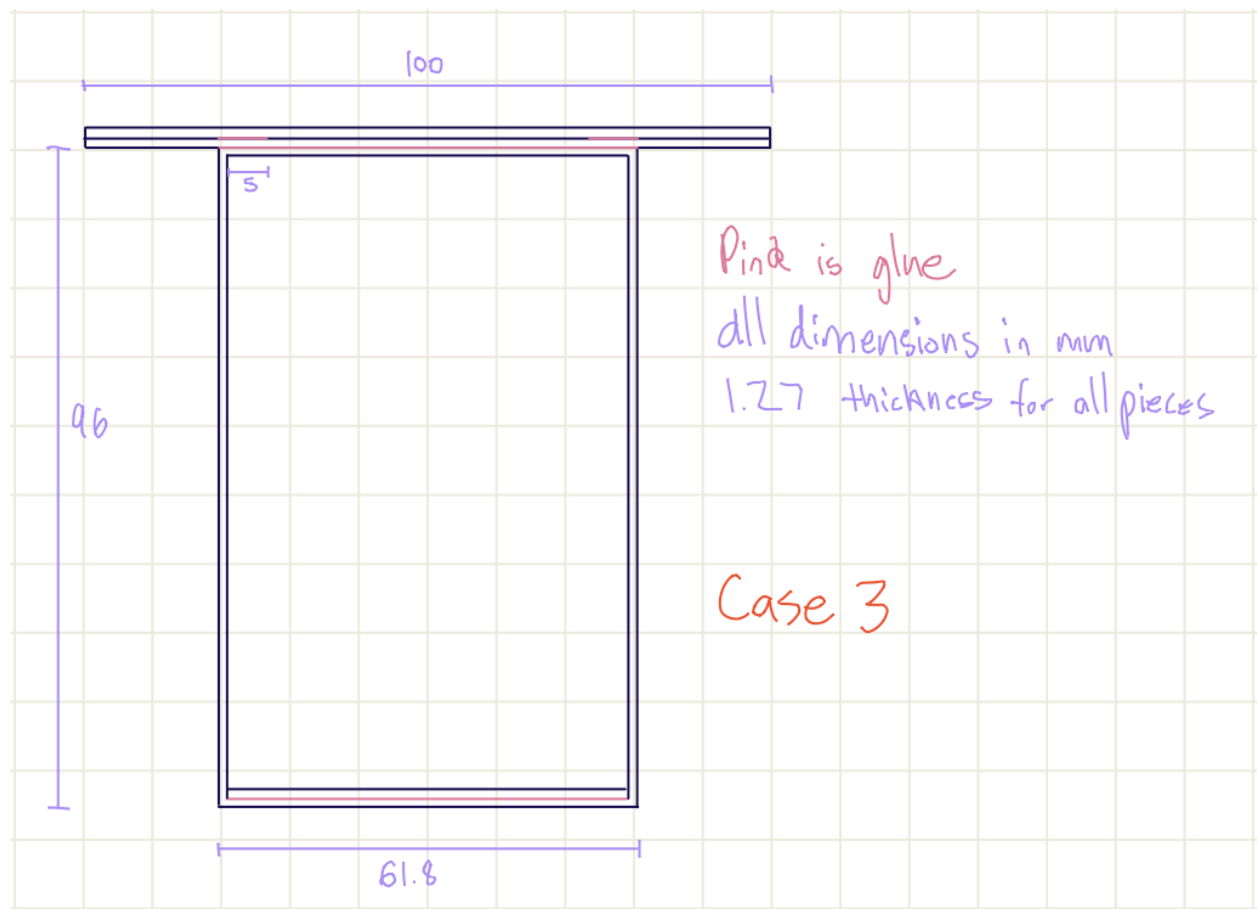


Figure 6: Cross Section Case 3 Diagram. This case is for distances in mm (450, 540) and (720, 810) along the bridge's span.

3.1.4. Cross Section Case 4: Yes O-Shape, No Double Bottom

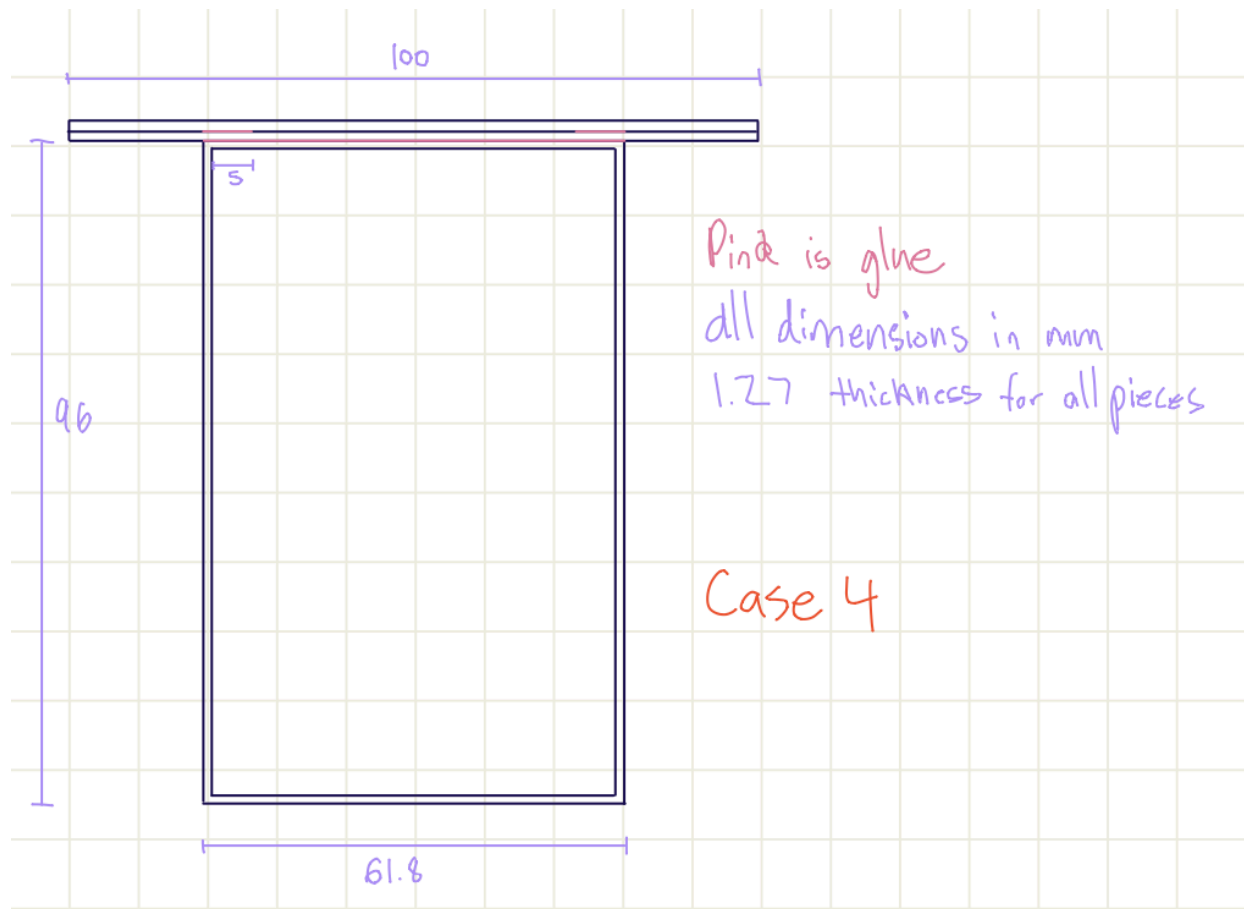


Figure 7: Cross Section Case 4 Diagram. This case is for the distance in mm (540, 720) along the bridge's span.

Table 3: Intermediate Calculations for Final Design's Cross Section Case 1 and Case 2 under Load Case 2. There is no second layer of the bottom flange for cross-section case 1, so there is no shear stress either. The significant figures in the table below are slightly inaccurate because MATLAB does not display trailing zeros. For example, thin plate buckling stress in the webs for cross-section case 2 has a value of 22.0 MPa, which is shown as 22 MPa, despite requiring three significant figures.

	Intermediate Calculation	Cross Section Case 1	Cross Section Case 2
1	Centroidal Axis Location From Bottom (mm)	56.8	57.2
2	Second Moment of Area (mm ⁴)	1126000	1105000
3	Top Bending Stress - Compression (MPa)	3.15	3.17
4	Bottom Bending Stress - Tension (MPa)	4.28	4.39
5	First Moment of Area at the Centroid (mm ³)	12730	12570
6	Shear Stress at the Centroid (MPa)	1.383	1.392
7	First Moment of Area at the Glue Tab (mm ³)	10280	10180
8	Shear Stress at the Glue Tab (MPa)	0.226	0.228
9	Shear Buckling Stress (MPa)	3.34	3.42
10	Thin Plate Buckling Stress - Top Flange - Middle (MPa)	24.1	24.1
11	Thin Plate Buckling Stress - Top Flange - Sides (MPa)	24.1	24.1
12	Thin Plate Buckling Stress - Webs (MPa)	21.6	22
13	Shear Stress Between Layers of the Top Flange (MPa)	0.1097	0.1107
14	Shear Stress Between Layers of the Bottom Flange (...)		0.0204

Table 4: Intermediate Calculations for Final Design's Cross Section Case 3 and Case 4 under Load Case 2. There is no second layer of the bottom flange for cross-section case 4, so there is no shear stress either. The significant figures in the table below are slightly inaccurate because MATLAB does not display trailing zeros. For example, shear stress in the glue tabs for cross-section case 4 has a value of 22.0 MPa, which is shown as 22 MPa, despite requiring three significant figures.

	Intermediate Calculation	Cross Section Case 3	Cross Section Case 4
1	Centroidal Axis Location From Bottom (mm)	60.2	59.8
2	Second Moment of Area (mm ⁴)	1181000	1204000
3	Top Bending Stress - Compression (MPa)	2.75	2.73
4	Bottom Bending Stress - Tension (MPa)	4.33	4.22
5	First Moment of Area at the Centroid (mm ³)	13480	13640
6	Shear Stress at the Centroid (MPa)	1.396	1.387
7	First Moment of Area at the Glue Tab (mm ³)	9410	9510
8	Shear Stress at the Glue Tab (MPa)	0.0433	0.043
9	Shear Buckling Stress (MPa)	3.42	3.34
10	Thin Plate Buckling Stress - Top Flange - Middle (MPa)	24.1	24.1
11	Thin Plate Buckling Stress - Top Flange - Sides (MPa)	24.1	24.1
12	Thin Plate Buckling Stress - Webs (MPa)	25.9	25.3
13	Shear Stress Between Layers of the Top Flange (MPa)	0.096	0.0952
14	Shear Stress Between Layers of the Bottom Flange (...)	0.0201	

3.2. Shear Force Capacities

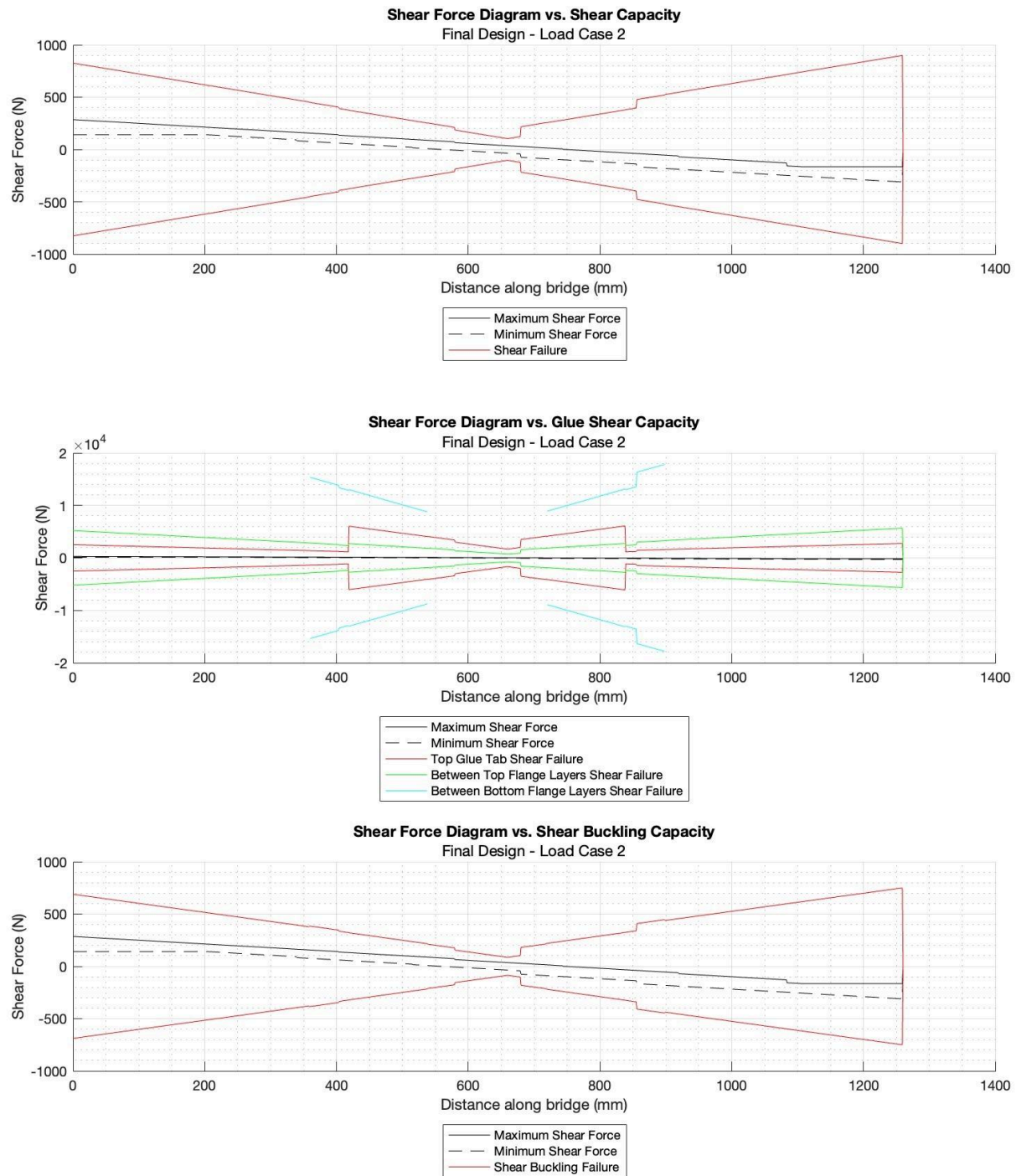


Figure 8: Shear Force Diagram vs. Shear Buckling Capacities for Final Design under Load Case 2. Closest to shear buckling failure. Decently far away from all other shear force failure modes. There is no second layer of the bottom flange for some lengths of the bridge, so there is no capacity for failure by shear stress in the glue between layers.

3.3 Bending Moment Capacities

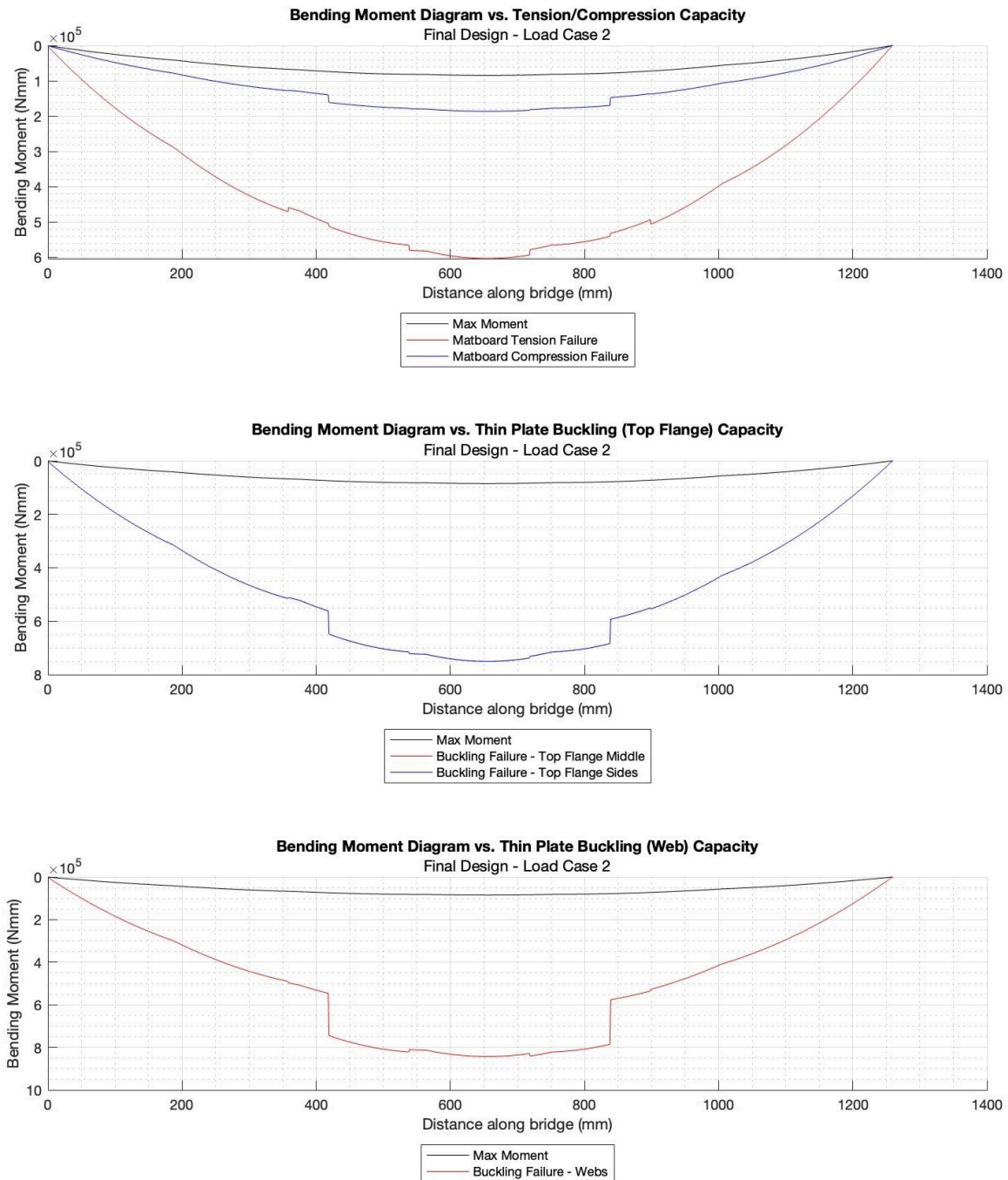


Figure 9: Bending Moment Diagram vs. Bending Moment Capacities for Final Design under Load Case 2. Closest to failing by compression. Significantly far from failing by other bending moment failure modes.

3.4 Factors of Safety & Maximum Train Weight

Table 5: Factors of safety against the previous failure modes in Section 3.2. and 3.3. Closest to failing by compression. This occurs at a maximum train weight of 854 N. The significant figures in the table below are slightly inaccurate because MATLAB does not display trailing zeros. For example, thin plate buckling in the webs has a FOS of 7.30, which is shown as 7.3, despite requiring three significant figures.

	Failure Mode	FOS
1	Web Shear Failure	2.87
2	Top Glue Tab Shear Failure	8.76
3	Shear Buckling Failure	2.41
4	Tension Failure	6.83
5	Compression Failure	1.89
6	Top Flange Middle Thin Plate Buckling Failure	7.61
7	Top Flange Sides Thin Plate Buckling Failure	7.6
8	Web Thin Plate Buckling Failure	7.3
9	Between Top Flange Layers Shear Failure	18.07
10	Between Bottom Flange Layers Shear Failure (if app...	98.2
11		
12	Maximum Train Weight (N)	854

Appendices

Appendix A: MATLAB Script for Section 2

```
clear; close all;
%% Basic Bridge Parameters
L = 1200; % Length of the bridge
n = 1200; % Discretize into 1 mm segments
P = 400; % Total weight of the train [N]
x = linspace(0, L, n+1); % x-axis
load_locations = [0 176 340 516 680 856]; % train load locations
%% Loading Conditions
point_loads = [1 1 1 1 1 1] * (P/6);
%% SFD + BMD
num_locations = L - load_locations(6) + 1; % number of train start locations
SFDi = zeros(num_locations, n+1); % 1 SFD for each train start location
BMDi = zeros(num_locations, n+1); % 1 BMD for each train start location
for i = 1:num_locations
    adjusted_load_locations = load_locations + i; % adjusted train load
    locations
    % sum of moments at A equation, rearranged for R_y (right support force)
    R_y = (sum(point_loads .* (adjusted_load_locations - 1))) / L;
    % sum of F_y equation, rearranged for L_y (left support force)
    L_y = sum(point_loads) - R_y;
    % Calculate shear along the bridge
    V = L_y; % initial shear force at left support
    SFDi(i, 1:adjusted_load_locations(1)) = V; % filling in a shear force value
    for a given range of x-values
        V = V - point_loads(1); % updating the shear force
        SFDi(i, (adjusted_load_locations(1)):adjusted_load_locations(2)) = V; %
        filling in the updated value for a given range of x-values
        V = V - point_loads(2); % repeat
        SFDi(i, adjusted_load_locations(2):adjusted_load_locations(3)) = V;
        V = V - point_loads(3);
        SFDi(i, adjusted_load_locations(3):adjusted_load_locations(4)) = V;
        V = V - point_loads(4);
        SFDi(i, adjusted_load_locations(4):adjusted_load_locations(5)) = V;
        V = V - point_loads(5);
        SFDi(i, adjusted_load_locations(5):adjusted_load_locations(6)) = V;
        V = V - point_loads(6);
        SFDi(i, adjusted_load_locations(6):n) = V;
        V = V + R_y; % final shear force update at right support
        SFDi(i, n+1) = V;
        BMDi(i,:) = cumsum(SFDi(i,:)); % integrating SFDi
    end
SFE = max(abs(SFDi)); % the absolute maximum values of shear force at all
locations
BME = max(BMDi); % the maximum value of all bending moments at all
locations
V_max = max(SFE); % max shear
M_max = max(BME); % max moment
```

```

%{
% Determine splice location
figure(3);
hold on;
normalized_SFE = (SFE - min(SFE)) / (max(SFE) + min(SFE));
normalized_BME = (BME - min(BME)) / (max(BME) + min(BME));
plot(x, normalized_SFE, "r");
plot(x, normalized_BME, "b");
plot(x, normalized_BME + normalized_SFE, "k");
%}

%% Define Cross Section Parameters
%% CHANGE THE VALUES HERE DURIGN DESIGN ITERATIONS
%% !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
% Top Flange Parameters
b_top_flange = zeros(n+1, 1);
t_top_flange = zeros(n+1, 1);
for i = 1:n+1
    b_top_flange(i) = 100;
    t_top_flange(i) = 1.27;
end
% Bot Flange Parameters
b_bot_flange = zeros(n+1, 1);
t_bot_flange = zeros(n+1, 1);
for i = 1:n+1
    b_bot_flange(i) = 80;
    t_bot_flange(i) = 1.27;
end
% Glue Tab Parameters
t_glue_tab = zeros(n+1, 1);
b_glue_tab = zeros(n+1, 1);
% Web Parameters
t_web = zeros(n+1, 1);
h_web = zeros(n+1, 1);
for i = 1:n+1
    t_glue_tab(i) = 1.27;
    t_web(i) = 1.27;
    h_web(i) = 75 - t_bot_flange(i) - t_glue_tab(i);
    b_glue_tab(i) = t_web(i) + 5;
end
%% Component Areas and Centroids
% centroids are calculated from the bottom of the cross-section
% Top Flange
top_flange_area = b_top_flange .* t_top_flange;
top_flange_centroid = t_top_flange / 2 + t_glue_tab + h_web + t_bot_flange;
% Bottom Flange
bot_flange_area = b_bot_flange .* t_top_flange;
bot_flange_centroid = t_bot_flange / 2;
% Web
web_area = t_web .* h_web;

```

```

web_centroid = h_web / 2 + t_bot_flange;
% Glue Tab
glue_tab_area = t_glue_tab .* b_glue_tab;
glue_tab_centroid = t_glue_tab / 2 + h_web + t_bot_flange;
%% Total Height
total_height = t_top_flange + t_glue_tab + h_web + t_bot_flange;
%% Cross Section Calculations
y_bar = zeros(n+1,1); % global centroid from the bottom
I = zeros(n+1,1); % second moment of area
bending_stress_top = zeros(n+1,1);
bending_stress_bot = zeros(n+1,1);
for i = 1:n+1
    %% Y_bar
    y_bar(i) = ((top_flange_area(i) * top_flange_centroid(i))...
        + (2 * glue_tab_area(i) * glue_tab_centroid(i))...
        + (2 * web_area(i) * web_centroid(i))...
        + (bot_flange_area(i) * bot_flange_centroid(i)))...
        / (top_flange_area(i) + bot_flange_area(i) + 2 * web_area(i) + 2 *
glue_tab_area(i));
    %% I
    I(i) = (top_flange_area(i) * (top_flange_centroid(i) - y_bar(i))^2)...
        + 2 * (glue_tab_area(i) * (glue_tab_centroid(i) - y_bar(i))^2)...
        + 2 * (web_area(i) * (web_centroid(i) - y_bar(i))^2)...
        + (bot_flange_area(i) * (bot_flange_centroid(i) - y_bar(i))^2)...
        + ((b_top_flange(i) * (t_top_flange(i)^3))/12)...
        + (2 * (b_glue_tab(i) * (t_glue_tab(i)^3))/12)...
        + (2 * (t_web(i) * (h_web(i)^3))/12)...
        + (b_bot_flange(i) * (t_bot_flange(i)^3))/12;
    %% Top Bending Stress
    bending_stress_top(i) = (M_max * (total_height(i) - y_bar(i)))/I(i);
    %% Bottom Bending Stress
    bending_stress_bot(i) = (M_max * y_bar(i))/I(i);
end
%% Shear Stresses
shear_stress_web = zeros(n+1,1);
Q_web = zeros(n+1, 1);
shear_stress_glue_tab = zeros(n+1, 1);
Q_glue_tab = zeros(n+1, 1);
for i = 1:n+1
    %% Q at the Web (Centroid)
    Q_web(i) = (2 * (y_bar(i) - t_bot_flange(i)) * t_web(i)) * ((y_bar(i) -
t_bot_flange(i))/2)...
        + bot_flange_area(i) * (y_bar(i) - bot_flange_centroid(i));
    %% Shear Stress at the Web (Centroid)
    shear_stress_web(i) = (V_max * Q_web(i))/(I(i) * 2 * t_web(i));
    %% Q at the Top Glue Tab
    Q_glue_tab(i) = top_flange_area(i) * (top_flange_centroid(i) - y_bar(i));
    %% Shear Stress at the Glue Tab

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```

    shear_stress_glue_tab(i) = (V_max * Q_glue_tab(i))/(I(i) * 2 *
b_glue_tab(i));
end
%% Material Properties
tensile_strength = 30;
compressive_strength = 6;
shear_strength = 4;
glue_shear_strength = 2;
poisson = 0.2;
youngs = 4000;
%% Shear Buckling
a = 400; % distance between diaphragms
shear_stress_buckle = ((5 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
    * ((t_web./h_web).^2 + (t_web./a).^2);
%% Thin Plate Buckling
% Case 1: Top Flange - Middle
tpb_tf_mid_stress = zeros(n+1, 1);
b_in_case1 = b_bot_flange - 2 * t_web/2;
for i = 1:n+1
    tpb_tf_mid_stress(i) = ((4 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
        * ((t_top_flange(i)/b_in_case1(i))^2);
end
% Case 2: Top Flange - Sides
tpb_tf_sides_stress = zeros(n+1, 1);
b_in_case2 = (b_top_flange - b_bot_flange) / 2 + t_web/2;
for i = 1:n+1
    tpb_tf_sides_stress(i) = ((0.4254 * pi^2 * youngs)/(12 * (1 -
poisson^2)))...
        * ((t_top_flange(i)/b_in_case2(i))^2);
end
% Case 3: Webs
tpb_webs_stress = zeros(n+1, 1);
y_web = zeros(n+1, 1);
b_web = t_web;
bending_stress_web = zeros(n+1, 1);
for i = 1:n+1
    y_web(i) = total_height(i) - y_bar(i) - t_top_flange(i);
    tpb_webs_stress(i) = ((6 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
        * ((b_web(i)/y_web(i))^2);
    bending_stress_web(i) = (M_max * y_web(i))/I(i);
end
%% Intermediate Calculations Table
format shortG;
intermediate = ["Centroidal Axis Location From Bottom (mm)"; "Second Moment of
Area (mm^4)"...
; "Top Bending Stress - Compression (MPa)"; "Bottom Bending Stress - Tension
(MPa)"...
; "First Moment of Area at the Centroid (mm^3)"; "Shear Stress at the Centroid
(MPa)"...

```



```

;"First Moment of Area at the Glue Tab (mm^3)"; "Shear Stress at the Glue Tab (MPa)"...
;"Shear Buckling Stress (MPa)"; "Thin Plate Buckling Stress - Top Flange - Middle (MPa)"...
;"Thin Plate Buckling Stress - Top Flange - Sides (MPa)"; "Thin Plate Buckling Stress - Webs (MPa)"];
values = [round(y_bar(1), 3, "significant")...
    ; round(I(1), 3, "significant")...
    ; round(bending_stress_top(1), 3, "significant")...
    ; round(bending_stress_bot(1), 3, "significant")...
    ; round(Q_web(1), 3, "significant")...
    ; round(shear_stress_web(1), 4, "significant")...
    ; round(Q_glue_tab(1), 3, "significant")...
    ; round(shear_stress_glue_tab(1), 3, "significant")...
    ; round(shear_stress_buckle(1), 3, "significant")...
    ; round(tpb_tf_mid_stress(1), 3, "significant")...
    ; round(tpb_tf_sides_stress(1), 3, "significant")...
    ; round(tpb_webs_stress(1), 3, "significant")];
intermediate_result_table = table(intermediate, values);
f = uifigure('Name', 'Intermediate Calculations Table'); % Create a UI figure
uitable(f, 'Data', intermediate_result_table{:, :}, ... % Extract data from the table
    'ColumnName', ["Intermediate Calculation", "Value"], ... % Set column names
    'Units', 'normalized', 'Position', [0, 0, 1, 1]);

%% FOS
FOS_tension = zeros(n+1, 1);
FOS_compression = zeros(n+1, 1);
FOS_shear = zeros(n+1, 1);
FOS_shear_glue_tab = zeros(n+1, 1);
FOS_shear_buckle = zeros(n+1, 1);
FOS_shear_tpb_tf_mid = zeros(n+1, 1);
FOS_shear_tpb_tf_sides = zeros(n+1, 1);
FOS_shear_tpb_web = zeros(n+1, 1);
for i = 1:n+1
    FOS_tension(i) = tensile_strength / bending_stress_bot(i);
    FOS_compression(i) = compressive_strength / bending_stress_top(i);
    FOS_shear(i) = shear_strength / shear_stress_web(i);
    FOS_shear_glue_tab(i) = glue_shear_strength / shear_stress_glue_tab(i);
    FOS_shear_buckle(i) = shear_stress_buckle(i) ./ shear_stress_web(i);
    FOS_shear_tpb_tf_mid(i) = tpb_tf_mid_stress(i) / bending_stress_top(i);
    FOS_shear_tpb_tf_sides(i) = tpb_tf_sides_stress(i) / bending_stress_top(i);
    FOS_shear_tpb_web(i) = tpb_webs_stress(i) / bending_stress_web(i);
end
% Factor of Safety Table
FailureMode = ["Web Shear Failure"...
    ; "Top Glue Tab Shear Failure"...
    ; "Shear Buckling Failure"...
    ; "Tension Failure"...

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    ;"Compression Failure"...
    ;"Top Flange Middle Thin Plate Buckling Failure"...
    ;"Top Flange Sides Thin Plate Buckling Failure"...
    ;"Web Thin Plate Buckling Failure"; ""; "Maximum Train Weight (N)"];
FOS = [round(min(FOS_shear), 3, "significant")...
    ;round(min(FOS_shear_glue_tab), 3, "significant")...
    ;round(min(FOS_shear_buckle), 3, "significant")...
    ;round(min(FOS_tension), 3, "significant")...
    ;round(min(FOS_compression), 4, "significant")...
    ;round(min(FOS_shear_tpb_tf_mid), 3, "significant")...
    ;round(min(FOS_shear_tpb_tf_sides), 3, "significant")...
    ;round(min(FOS_shear_tpb_web), 3, "significant")];
FOS_min = min(FOS);
FOS = [FOS; NaN; round(FOS_min * P, 3, "significant")];
FOS_result_table = table(FailureMode, FOS);
% formatting
f = uifigure('Name', 'Factor of Safety Table'); % Create a UI figure
t = uitable(f, 'Data', FOS_result_table{:, :}, ... % Extract data from the
table
    'ColumnName', ["Failure Mode", "FOS"], ... % Set column names
    'Units', 'normalized', 'Position', [0, 0, 1, 1]); % Set the position
and size of the table in the figure window
%% PLOTS
figure(2);
set(gcf, 'Units', 'Normalized', 'OuterPosition', [0 0 1 1]);
tiledlayout(3, 2, 'TileSpacing', 'Compact', 'Padding', 'Compact');
design = "Design 0 - Load Case 1";
%% Plot 1 - Matboard Web Shear Failure
nexttile;
hold on; grid on; grid minor;
title("Shear Force Diagram vs. Shear Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear' .* max(abs(SFDi)), 'r', DisplayName="Shear Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear' .* -max(abs(SFDi)), 'r', DisplayName="");
%% Plot 2 - Matboard Tension/Compression Failure
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Tension/Compression Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')
legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_tension' .* max(BMDi), 'r', DisplayName="Matboard Tension Failure")

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```

plot(x, FOS_compression' .* max(BMDi), 'b', DisplayName="Matboard Compression Failure")
%% Plot 3 - Glue Shear Failure
nexttile;
title("Shear Force Diagram vs. Glue Shear Capacity", design)
hold on; grid on; grid minor;
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear_glue_tab' .* max(abs(SFDi)), 'r', DisplayName="Top Glue Tab Shear Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear_glue_tab' .* -max(abs(SFDi)), 'r', DisplayName="");
%% Plot 4 - Matboard TPB - Top Flange Failure
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Thin Plate Buckling (Top Flange) Capacity", design);
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')
legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_shear_tpb_tf_mid' .* max(BMDi), 'r', DisplayName="Buckling Failure - Top Flange Middle")
plot(x, FOS_shear_tpb_tf_sides' .* max(BMDi), 'b', DisplayName="Buckling Failure - Top Flange Sides")
%% Plot 5 - Shear Buckling Failure
nexttile;
hold on; grid on; grid minor;
title("Shear Force Diagram vs. Shear Buckling Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear_buckle' .* max(abs(SFDi)), 'r', DisplayName="Shear Buckling Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear_buckle' .* -max(abs(SFDi)), 'r', DisplayName="");
%% Plot 6 - Matboard TPB - Web
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Thin Plate Buckling (Web) Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')

```

```

legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_shear_tpb_web' .* max(BMDi), 'r', DisplayName="Buckling Failure - Webs")

```

Appendix B: MATLAB Script for Section 3

```

clear; close all;
%% Basic Bridge Parameters
L = 1260; % Length of the bridge
n = 1260; % Discretize into 1 mm segments
% Load Case 2 - Base Case
P = 135 * 2 + 182;
% Variable Load Case
% lightest_car = 135;
% P = lightest_car + 1.1 * lightest_car + 1.1 * 1.35 * lightest_car; % Total
weight of the train [N]
x = linspace(0, L, n+1); % x-axis
load_locations = [0 176 340 516 680 856]; % train load locations
%% Loading Conditions
% Load Case 2 - Base Case
point_loads = [135/2, 135/2, 135/2, 135/2, 182/2, 182/2];
% Variable Load Case
% point_loads = [(1.1 * lightest_car)/2, ((1.1 * lightest_car)/2)...
%      , lightest_car/2, lightest_car/2, (1.1 * 1.35 * lightest_car)/2, (1.1 *
1.35 * lightest_car)/2];
%% SFD + BMD
num_locations = L - load_locations(6) + 1; % number of train start locations
SFDi = zeros(num_locations, n+1); % 1 SFD for each train start location
BMDi = zeros(num_locations, n+1); % 1 BMD for each train start location
for i = 1:num_locations
    adjusted_load_locations = load_locations + i; % adjusted train load
locations
    % sum of moments at A equation, rearranged for R_y (right support force)
    R_y = (sum(point_loads .* (adjusted_load_locations - 1))) / L;
    % sum of F_y equation, rearranged for L_y (left support force)
    L_y = sum(point_loads) - R_y;
    % Calculate shear along the bridge
    V = L_y; % initial shear force at left support
    SFDi(i, 1:adjusted_load_locations(1)) = V; % filling in a shear force value
for a given range of x-values
    V = V - point_loads(1); % updating the shear force
    SFDi(i, (adjusted_load_locations(1)):adjusted_load_locations(2)) = V; %
filling in the updated value for a given range of x-values
    V = V - point_loads(2); % repeat
    SFDi(i, adjusted_load_locations(2):adjusted_load_locations(3)) = V;
    V = V - point_loads(3);
    SFDi(i, adjusted_load_locations(3):adjusted_load_locations(4)) = V;
    V = V - point_loads(4);
    SFDi(i, adjusted_load_locations(4):adjusted_load_locations(5)) = V;

```

```

V = V - point_loads(5);
SFDi(i, adjusted_load_locations(5):adjusted_load_locations(6)) = V;
V = V - point_loads(6);
SFDi(i, adjusted_load_locations(6):n) = V;
V = V + R_y; % final shear force update at right support
SFDi(i, n+1) = V;
BMDi(i,:) = cumsum(SFDi(i,:)); % integrating SFDi
end
SFE = max(abs(SFDi)); % the absolute maximum values of shear force at all
locations
BME = max(BMDi); % the maximum value of all bending moments at all
locations
V_max = max(SFE); % max shear
M_max = max(BME); % max moment
%{
% Determine splice location
figure(3);
hold on;
normalized_SFE = (SFE - min(SFE)) / (max(SFE) + min(SFE));
normalized_BME = (BME - min(BME)) / (max(BME) + min(BME));
plot(x, normalized_SFE, "r");
plot(x, normalized_BME, "b");
plot(x, normalized_BME + normalized_SFE, "k");
%}
%% Define Cross Section Parameters
%% CHANGE THE VALUES HERE DURIGN DESIGN ITERATIONS
%% !!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!
% Top Flange Parameters
b_top_flange = zeros(n+1, 1);
t_top_flange = zeros(n+1, 1);
for i = 1:n+1
    b_top_flange(i) = 100;
    t_top_flange(i) = 1.27 * 2; % there will be glue shear inbetween
end
% Bot Flange Parameters
b_bot_flange = zeros(n+1, 1);
t_bot_flange = zeros(n+1, 1);
for i = 1:n+1
    b_bot_flange(i) = 61.79; % optimized via math
    if i < 180 * 2 % between diaphragms
        t_bot_flange(i) = 1.27;
    elseif i < 180 * 3
        t_bot_flange(i) = 1.27 * 2;
    elseif i < 180 * 4
        t_bot_flange(i) = 1.27;
    elseif i < 180 * 5
        t_bot_flange(i) = 1.27 * 2;
    else
        t_bot_flange(i) = 1.27;
    end
end

```

```

    end
end
% Glue Tab Parameters
t_glue_tab = zeros(n+1, 1);
b_glue_tab = zeros(n+1, 1);
% Web Parameters
t_web = zeros(n+1, 1);
h_web = zeros(n+1, 1);
for i = 1:n+1
    t_glue_tab(i) = 1.27;
    t_web(i) = 1.27;
    h_web(i) = 96 - t_bot_flange(i) - t_glue_tab(i);
    if i < L / 3
        b_glue_tab(i) = t_web(i) + 5;
    elseif i < 2 * L / 3
        b_glue_tab(i) = t_web(i) + 27.315; % o shape in the centre
    else
        b_glue_tab(i) = t_web(i) + 5;
    end
end
end
%% Component Areas and Centroids
% centroids are calculated from the bottom of the cross-section
% Top Flange
top_flange_area = b_top_flange .* t_top_flange;
top_flange_centroid = t_top_flange / 2 + t_glue_tab + h_web + t_bot_flange;
% Bottom Flange
bot_flange_area = b_bot_flange .* t_top_flange;
bot_flange_centroid = t_bot_flange / 2;
% Web
web_area = t_web .* h_web;
web_centroid = h_web / 2 + t_bot_flange;
% Glue Tab
glue_tab_area = t_glue_tab .* b_glue_tab;
glue_tab_centroid = t_glue_tab / 2 + h_web + t_bot_flange;
%% Total Height
total_height = t_top_flange + t_glue_tab + h_web + t_bot_flange;
%% Cross Section Calculations
y_bar = zeros(n+1,1); % global centroid from the bottom
I = zeros(n+1,1); % second moment of area
bending_stress_top = zeros(n+1,1);
bending_stress_bot = zeros(n+1,1);
for i = 1:n+1
    %% Y_bar
    y_bar(i) = ((top_flange_area(i) * top_flange_centroid(i))...
        + (2 * glue_tab_area(i) * glue_tab_centroid(i))...
        + (2 * web_area(i) * web_centroid(i))...
        + (bot_flange_area(i) * bot_flange_centroid(i)))...
        / (top_flange_area(i) + bot_flange_area(i) + 2 * web_area(i) + 2 *
glue_tab_area(i));

```

```

%% I
I(i) = (top_flange_area(i) * (top_flange_centroid(i) - y_bar(i))^2)...
      + 2 * (glue_tab_area(i) * (glue_tab_centroid(i) - y_bar(i))^2)...
      + 2 * (web_area(i) * (web_centroid(i) - y_bar(i))^2)...
      + (bot_flange_area(i) * (bot_flange_centroid(i) - y_bar(i))^2)...
      + ((b_top_flange(i) * (t_top_flange(i)^3))/12)...
      + (2 * (b_glue_tab(i) * (t_glue_tab(i)^3))/12)...
      + (2 * (t_web(i) * (h_web(i)^3))/12)...
      + (b_bot_flange(i) * (t_bot_flange(i)^3))/12;
%% Top Bending Stress
bending_stress_top(i) = (M_max * (total_height(i) - y_bar(i)))/I(i);
%% Bottom Bending Stress
bending_stress_bot(i) = (M_max * y_bar(i))/I(i);
end
%% Shear Stresses
shear_stress_web = zeros(n+1,1);
Q_web = zeros(n+1, 1);
shear_stress_glue_tab = zeros(n+1, 1);
Q_glue_tab = zeros(n+1, 1);
Q_between_top_flange = zeros(n+1, 1);
shear_stress_between_top = zeros(n+1, 1);
Q_between_bot_flange = NaN(n+1, 1); % exists for some of it
shear_stress_between_bot = NaN(n+1, 1);
for i = 1:n+1
    %% Q at the Web (Centroid)
    Q_web(i) = (2 * (y_bar(i) - t_bot_flange(i)) * t_web(i)) * ((y_bar(i) -
t_bot_flange(i))/2)...
      + bot_flange_area(i) * (y_bar(i) - bot_flange_centroid(i));
    %% Shear Stress at the Web (Centroid)
    shear_stress_web(i) = (V_max * Q_web(i))/(I(i) * 2 * t_web(i));
    %% Q at the Top Glue Tab
    Q_glue_tab(i) = top_flange_area(i) * (top_flange_centroid(i) - y_bar(i));
    %% Shear Stress at the Glue Tab
    shear_stress_glue_tab(i) = (V_max * Q_glue_tab(i))/(I(i) * 2 *
b_glue_tab(i));
    % Q between top flange layers
    Q_between_top_flange(i) = top_flange_area(i)/2 * (top_flange_centroid(i) +
t_top_flange(i)/2 - y_bar(i));
    shear_stress_between_top(i) = (V_max * Q_between_top_flange(i))/(I(i) * 2 *
6.67); % manually calculated for simplicity
    % Q between bot flange layers
    if ((i > 180 * 2) && (i < 180 * 3)) || ((i > 180 * 4) && (i < 180 * 5))
        Q_between_bot_flange(i) = bot_flange_area(i)/2 * (y_bar(i) -
bot_flange_centroid(i) - t_bot_flange(i)/2);
        shear_stress_between_bot(i) = (V_max * Q_between_bot_flange(i))/(I(i) *
(b_bot_flange(i) - 2 * t_web(i)));
    elseif i < 180 * 5
        t_bot_flange(i) = 1.27 * 2;
    else

```

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        t_bot_flange(i) = 1.27;
    end
end
%% Material Properties
tensile_strength = 30;
compressive_strength = 6;
shear_strength = 4;
glue_shear_strength = 2;
poisson = 0.2;
youngs = 4000;
%% Shear Buckling
a = 400; % distance between diaphragms
shear_stress_buckle = ((5 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
    * ((t_web./h_web).^2 + (t_web./a).^2);
%% Thin Plate Buckling
% Case 1: Top Flange - Middle
tpb_tf_mid_stress = zeros(n+1, 1);
b_in_case1 = b_bot_flange - 2 * t_web/2;
for i = 1:n+1
    tpb_tf_mid_stress(i) = ((4 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
        * ((t_top_flange(i)/b_in_case1(i))^2);
end
% Case 2: Top Flange - Sides
tpb_tf_sides_stress = zeros(n+1, 1);
b_in_case2 = (b_top_flange - b_bot_flange) / 2 + t_web/2;
for i = 1:n+1
    tpb_tf_sides_stress(i) = ((0.4254 * pi^2 * youngs)/(12 * (1 -
poisson^2)))...
        * ((t_top_flange(i)/b_in_case2(i))^2);
end
% Case 3: Webs
tpb_webs_stress = zeros(n+1, 1);
y_web = zeros(n+1, 1);
b_web = t_web;
bending_stress_web = zeros(n+1, 1);
for i = 1:n+1
    y_web(i) = total_height(i) - y_bar(i) - t_top_flange(i);
    tpb_webs_stress(i) = ((6 * pi^2 * youngs)/(12 * (1 - poisson^2)))...
        * ((b_web(i)/y_web(i))^2);
    bending_stress_web(i) = (M_max * y_web(i))/I(i);
end
%% Intermediate Calculations Table(s)
% Cross Section 1 (No O, No Splice)
location1 = 1;
intermediate = ["Centroidal Axis Location From Bottom (mm)"; "Second Moment of
Area (mm^4)"...
    ; "Top Bending Stress - Compression (MPa)"; "Bottom Bending Stress - Tension
(MPa)"...

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    ;"First Moment of Area at the Centroid (mm^3)"; "Shear Stress at the
Centroid (MPa)"...
    ;"First Moment of Area at the Glue Tab (mm^3)"; "Shear Stress at the Glue
Tab (MPa)"...
    ;"Shear Buckling Stress (MPa)"; "Thin Plate Buckling Stress - Top Flange -
Middle (MPa)"...
    ;"Thin Plate Buckling Stress - Top Flange - Sides (MPa)"; "Thin Plate
Buckling Stress - Webs (MPa)"...
    ;"Shear Stress Between Layers of the Top Flange (MPa)"...
    ;"Shear Stress Between Layers of the Bottom Flange (MPa)"];
values1 = [round(y_bar(location1), 3, "significant")...
    ; round(I(location1), 4, "significant")...
    ; round(bending_stress_top(location1), 3, "significant")...
    ; round(bending_stress_bot(location1), 3, "significant")...
    ; round(Q_web(location1), 4, "significant")...
    ; round(shear_stress_web(location1), 4, "significant")...
    ; round(Q_glue_tab(location1), 4, "significant")...
    ; round(shear_stress_glue_tab(location1), 3, "significant")...
    ; round(shear_stress_buckle(location1), 3, "significant")...
    ; round(tpb_tf_mid_stress(location1), 3, "significant")...
    ; round(tpb_tf_sides_stress(location1), 3, "significant")...
    ; round(tpb_webs_stress(location1), 3, "significant")...
    ; round(shear_stress_between_top(location1), 4, "significant")...
    ; round(shear_stress_between_bot(location1), 3, "significant")];
% Cross Section 2 (No O, Yes Splice)
location2 = 180 * 2 + 10;
values2 = [round(y_bar(location2), 3, "significant")...
    ; round(I(location2), 4, "significant")...
    ; round(bending_stress_top(location2), 3, "significant")...
    ; round(bending_stress_bot(location2), 3, "significant")...
    ; round(Q_web(location2), 4, "significant")...
    ; round(shear_stress_web(location2), 4, "significant")...
    ; round(Q_glue_tab(location2), 4, "significant")...
    ; round(shear_stress_glue_tab(location2), 3, "significant")...
    ; round(shear_stress_buckle(location2), 3, "significant")...
    ; round(tpb_tf_mid_stress(location2), 3, "significant")...
    ; round(tpb_tf_sides_stress(location2), 3, "significant")...
    ; round(tpb_webs_stress(location2), 3, "significant")...
    ; round(shear_stress_between_top(location2), 4, "significant")...
    ; round(shear_stress_between_bot(location2), 3, "significant")];
% Cross Section 3 (Yes O, Yes Splice)
location3 = 180 * 3 - 10;
values3 = [round(y_bar(location3), 3, "significant")...
    ; round(I(location3), 4, "significant")...
    ; round(bending_stress_top(location3), 3, "significant")...
    ; round(bending_stress_bot(location3), 3, "significant")...
    ; round(Q_web(location3), 4, "significant")...
    ; round(shear_stress_web(location3), 4, "significant")...
    ; round(Q_glue_tab(location3), 3, "significant")...

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; round(shear_stress_glue_tab(location3), 3, "significant")...
; round(shear_stress_buckle(location3), 3, "significant")...
; round(tpb_tf_mid_stress(location3), 3, "significant")...
; round(tpb_tf_sides_stress(location3), 3, "significant")...
; round(tpb_webs_stress(location3), 3, "significant")...
; round(shear_stress_between_top(location3), 3, "significant")...
; round(shear_stress_between_bot(location3), 3, "significant")];
% Cross Section 3 (Yes O, No Splice)
location4 = 180 * 3 + 10;
values4 = [round(y_bar(location4), 3, "significant")...
; round(I(location4), 4, "significant")...
; round(bending_stress_top(location4), 3, "significant")...
; round(bending_stress_bot(location4), 3, "significant")...
; round(Q_web(location4), 4, "significant")...
; round(shear_stress_web(location4), 4, "significant")...
; round(Q_glue_tab(location4), 3, "significant")...
; round(shear_stress_glue_tab(location4), 3, "significant")...
; round(shear_stress_buckle(location4), 3, "significant")...
; round(tpb_tf_mid_stress(location4), 3, "significant")...
; round(tpb_tf_sides_stress(location4), 3, "significant")...
; round(tpb_webs_stress(location4), 3, "significant")...
; round(shear_stress_between_top(location4), 3, "significant")...
; round(shear_stress_between_bot(location4), 3, "significant")];
intermediate_result_table1 = table(intermediate, values1, values2);
f = uifigure('Name', 'Intermediate Calculations Table (CS1)'); % Create a UI
figure
uitable(f, 'Data', intermediate_result_table1{:, :}, ... % Extract data from
the table
'ColumnName', ["Intermediate Calculation", "Cross Section Case 1"...
, "Cross Section Case 2"], ... % Set column names
'Units', 'normalized', 'Position', [0, 0, 1, 1]);
intermediate_result_table2 = table(intermediate, values3, values4);
f = uifigure('Name', 'Intermediate Calculations Table (CS1)'); % Create a UI
figure
uitable(f, 'Data', intermediate_result_table2{:, :}, ... % Extract data from
the table
'ColumnName', ["Intermediate Calculation"...
, "Cross Section Case 3"...
, "Cross Section Case 4"], ... % Set column names
'Units', 'normalized', 'Position', [0, 0, 1, 1]);

%% FOS
FOS_tension = zeros(n+1, 1);
FOS_compression = zeros(n+1, 1);
FOS_shear = zeros(n+1, 1);
FOS_shear_glue_tab = zeros(n+1, 1);
FOS_shear_buckle = zeros(n+1, 1);
FOS_shear_tpb_tf_mid = zeros(n+1, 1);
FOS_shear_tpb_tf_sides = zeros(n+1, 1);
FOS_shear_tpb_web = zeros(n+1, 1);

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% New Values
FOS_shear_glue_between_top = zeros(n+1, 1);
FOS_shear_glue_between_bot = NaN(n+1, 1);
for i = 1:n+1
    FOS_tension(i) = tensile_strength / bending_stress_bot(i);
    FOS_compression(i) = compressive_strength / bending_stress_top(i);
    FOS_shear(i) = shear_strength / shear_stress_web(i);
    FOS_shear_glue_tab(i) = glue_shear_strength / shear_stress_glue_tab(i);
    FOS_shear_buckle(i) = shear_stress_buckle(i) ./ shear_stress_web(i);
    FOS_shear_tpb_tf_mid(i) = tpb_tf_mid_stress(i) / bending_stress_top(i);
    FOS_shear_tpb_tf_sides(i) = tpb_tf_sides_stress(i) / bending_stress_top(i);
    FOS_shear_tpb_web(i) = tpb_webs_stress(i) / bending_stress_web(i);
    FOS_shear_glue_between_top(i) = glue_shear_strength /
shear_stress_between_top(i);
    FOS_shear_glue_between_bot(i) = glue_shear_strength /
shear_stress_between_bot(i);
end
% Factor of Safety Table
FailureMode = ["Web Shear Failure"...
    ;"Top Glue Tab Shear Failure"...
    ;"Shear Buckling Failure"...
    ;"Tension Failure"...
    ;"Compression Failure"...
    ;"Top Flange Middle Thin Plate Buckling Failure"...
    ;"Top Flange Sides Thin Plate Buckling Failure"...
    ;"Web Thin Plate Buckling Failure"...
    ;"Between Top Flange Layers Shear Failure"...
    ;"Between Bottom Flange Layers Shear Failure (if applicable)"...
    ;""; "Maximum Train Weight (N)"];
FOS = [round(min(FOS_shear), 3, "significant")...
    ;round(min(FOS_shear_glue_tab), 3, "significant")...
    ;round(min(FOS_shear_buckle), 3, "significant")...
    ;round(min(FOS_tension), 3, "significant")...
    ;round(min(FOS_compression), 4, "significant")...
    ;round(min(FOS_shear_tpb_tf_mid), 3, "significant")...
    ;round(min(FOS_shear_tpb_tf_sides), 3, "significant")...
    ;round(min(FOS_shear_tpb_web), 3, "significant")...
    ;round(min(FOS_shear_glue_between_top), 4, "significant")...
    ;round(min(FOS_shear_glue_between_bot), 3, "significant")];
FOS_min = min(FOS);
FOS = [FOS; NaN; round(FOS_min * P, 3, "significant")];
FOS_result_table = table(FailureMode, FOS);
f = uifigure('Name', 'Factor of Safety Table'); % Create a UI figure
t = uitable(f, 'Data', FOS_result_table{:, :}, ... % Extract data from the
table
    'ColumnName', ["Failure Mode", "FOS"], ... % Set column names
    'Units', 'normalized', 'Position', [0, 0, 1, 1]); % Set the position
and size of the table in the figure window
%% PLOTS

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figure(2);
set(gcf, 'Units', 'Normalized', 'OuterPosition', [0 0 1 1]);
tiledlayout(3, 2, 'TileSpacing', 'Compact', 'Padding', 'Compact');
design = "Final Design - Load Case 2";
%% Plot 1 - Matboard Web Shear Failure
nexttile;
hold on; grid on; grid minor;
title("Shear Force Diagram vs. Shear Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear' .* max(abs(SFDi)), 'r', DisplayName="Shear Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear' .* -max(abs(SFDi)), 'r', DisplayName="");
%% Plot 2 - Matboard Tension/Compression Failure
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Tension/Compression Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')
legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_tension' .* max(BMDi), 'r', DisplayName="Matboard Tension Failure")
plot(x, FOS_compression' .* max(BMDi), 'b', DisplayName="Matboard Compression Failure")
%% Plot 3 - Glue Shear Failure
nexttile;
title("Shear Force Diagram vs. Glue Shear Capacity", design)
hold on; grid on; grid minor;
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear_glue_tab' .* max(abs(SFDi)), 'r', DisplayName="Top Glue Tab Shear Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear_glue_tab' .* -max(abs(SFDi)), 'r', DisplayName="");
legend(AutoUpdate="on");
plot(x, FOS_shear_glue_between_top' .* max(abs(SFDi)), 'b',
DisplayName="Between Top Flange Layers Shear Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear_glue_between_top' .* -max(abs(SFDi)), 'b', DisplayName="");
legend(AutoUpdate="on");
plot(x, FOS_shear_glue_between_bot' .* max(abs(SFDi)), 'c',
DisplayName="Between Bottom Flange Layers Shear Failure");

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legend(AutoUpdate="off");
plot(x, FOS_shear_glue_between_bot' .* -max(abs(SFDi)), 'c', DisplayName="");
%% Plot 4 - Matboard TPB - Top Flange Failure
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Thin Plate Buckling (Top Flange) Capacity",
design);
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')
legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_shear_tpb_tf_mid' .* max(BMDi), 'r', DisplayName="Buckling Failure
- Top Flange Middle")
plot(x, FOS_shear_tpb_tf_sides' .* max(BMDi), 'b', DisplayName="Buckling
Failure - Top Flange Sides")
%% Plot 5 - Shear Buckling Failure
nexttile;
hold on; grid on; grid minor;
title("Shear Force Diagram vs. Shear Buckling Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Shear Force (N)')
legend(Location="southoutside");
plot(x, max(SFDi), 'k', DisplayName="Maximum Shear Force");
plot(x, min(SFDi), '--k', DisplayName="Minimum Shear Force");
plot(x, FOS_shear_buckle' .* max(abs(SFDi)), 'r', DisplayName="Shear Buckling
Failure");
legend(AutoUpdate="off");
plot(x, FOS_shear_buckle' .* -max(abs(SFDi)), 'r', DisplayName="");
%% Plot 6 - Matboard TPB - Web
nexttile;
hold on; grid on; grid minor;
set(gca, 'YDir', 'reverse')
title("Bending Moment Diagram vs. Thin Plate Buckling (Web) Capacity", design)
xlabel('Distance along bridge (mm)')
ylabel('Bending Moment (Nmm)')
legend(Location="southoutside");
plot(x, BME, 'k', DisplayName="Max Moment")
plot(x, FOS_shear_tpb_web' .* max(BMDi), 'r', DisplayName="Buckling Failure -
Webs")

```